

STA302 Report

Group 173: Oscar Heath, Evelyn Hughes, Amy Ye

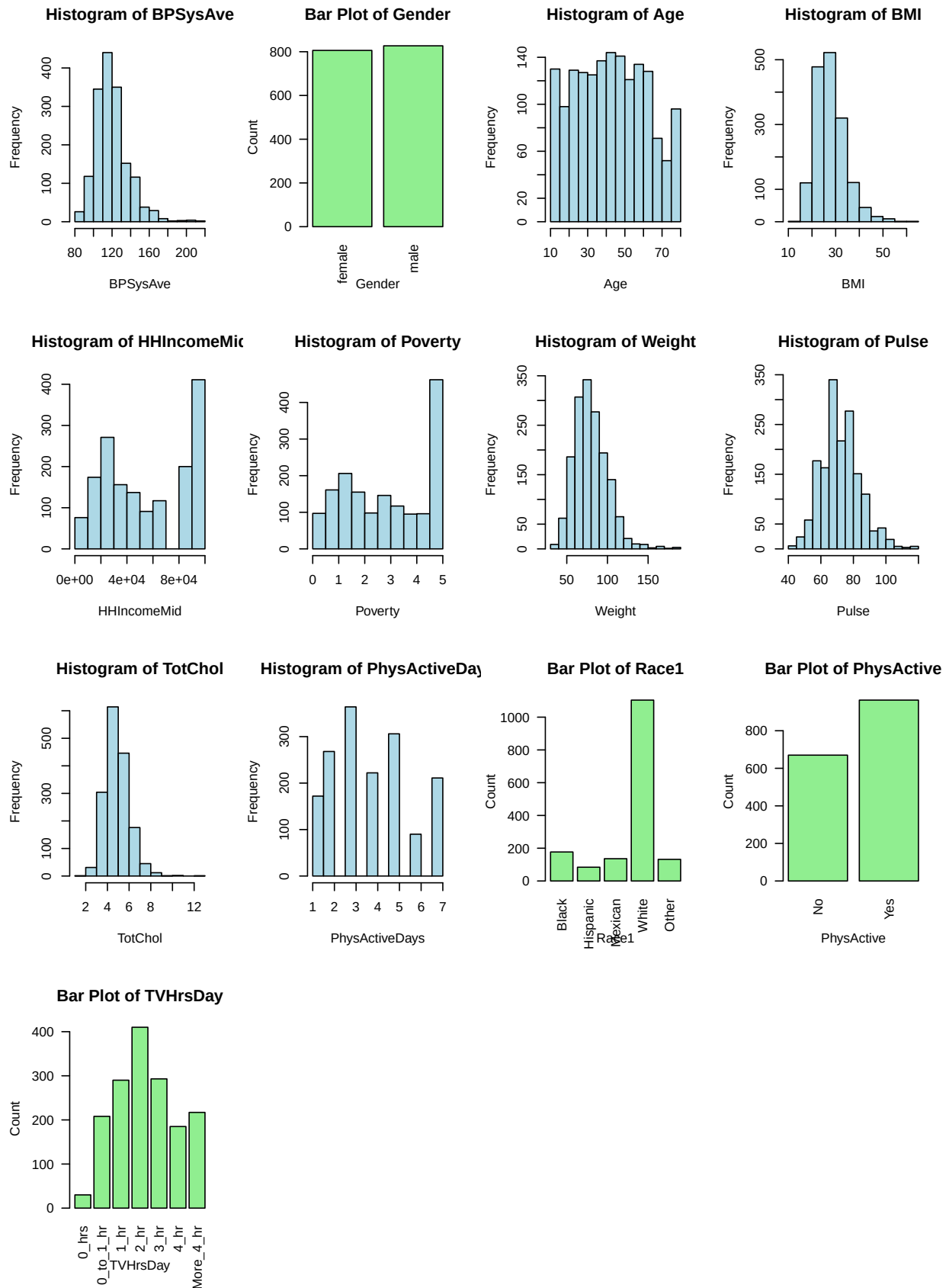
Importing Data from NHANES package

```
#import libraries  
library(NHANES)  
library(MASS)  
library(car)
```

```
## Loading required package: carData
```

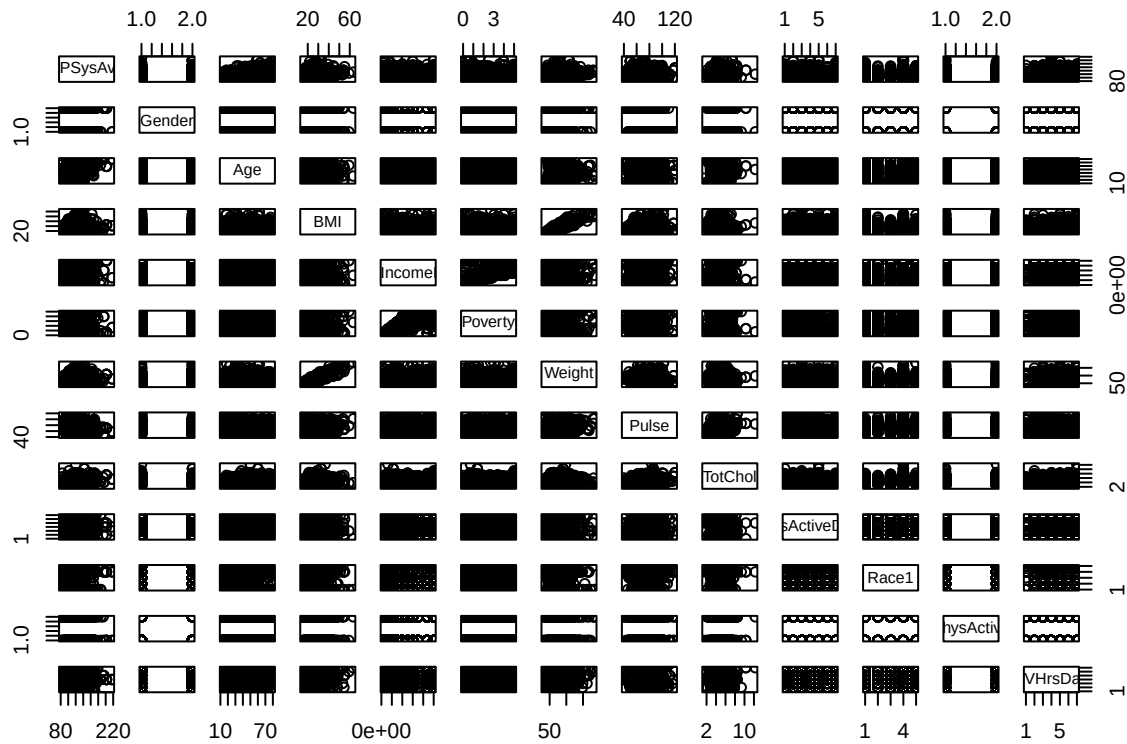
```
data(NHANES)  
  
# select columns we want from NHANES  
data <- subset(as.data.frame(NHANES),  
               select = c(BPSysAve, Gender, Age, BMI,  
                           HHIIncomeMid, Poverty, Weight, Pulse, TotChol, PhysActiveDays, Race1, PhysActiv  
data = na.omit(data)  
set.seed(42)  
s <- sample(1:nrow(data), 1000, replace=F)  
train <- data[s, ]  
test <- data[-s, ]
```

Plot the data to see distributions:



Checking correlation between predictors:

```
plot(data)
```



Summary of BPSysAve, our response variable

```
summary(data$BPSysAve)
```

```
##      Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      81.0   109.0   118.0   120.2   129.0   217.0
```

Graphing interaction terms:

```
par(mfrow = c(1, 1), mar = c(5, 4, 4, 2) + 0.1)

# Make a scatterplot of blood pressure and age, coloured by gender
plot(data$Age, data$BPSysAve,
     main = "Interaction of Age and Gender on Blood Pressure",
     xlab = "Age",
     ylab = "Systolic Blood Pressure",
```

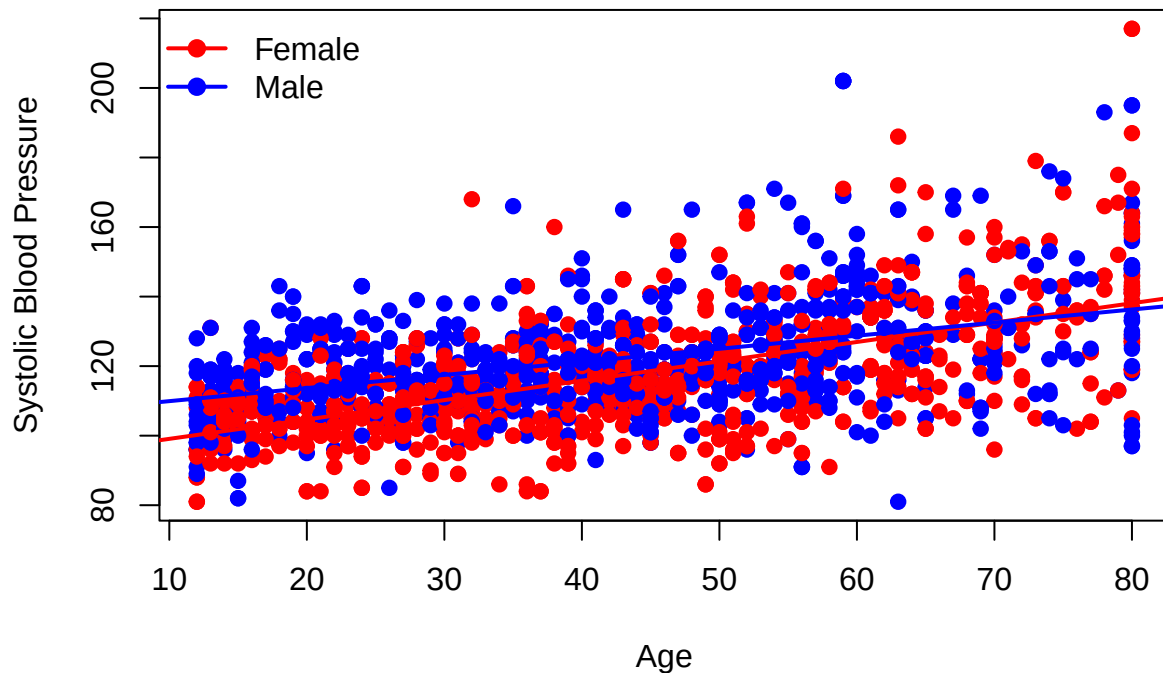
```

pch = 19,
col = ifelse(data$Gender == "male", "blue", "red"))
# Add female line of best fit
female_fit <- lm(BPSysAve ~ Age, data = subset(data, Gender == "female"))
abline(female_fit, col = "red", lwd = 2)
# Add male line of best fit
male_fit <- lm(BPSysAve ~ Age, data = subset(data, Gender == "male"))
abline(male_fit, col = "blue", lwd = 2)

legend("topleft",
      legend = c("Female", "Male"),
      col = c("red", "blue"),
      pch = 19, lwd = 2, bty = "n")

```

Interaction of Age and Gender on Blood Pressure



```

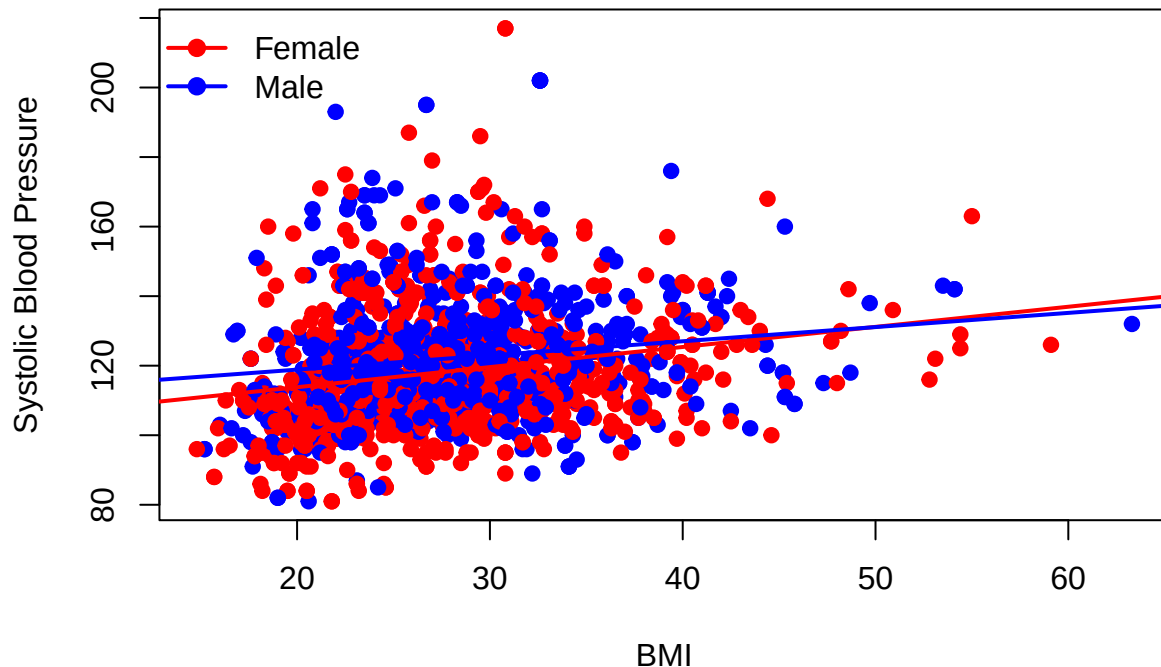
# Make a scatterplot of blood pressure and BMI
plot(data$BMI, data$BPSysAve,
      main = "Interaction of BMI and Gender on Blood Pressure",
      xlab = "BMI",
      ylab = "Systolic Blood Pressure",
      pch = 19,
      col = ifelse(data$Gender == "male", "blue", "red"))
# Add female line of best fit
female_fit_bmi <- lm(BPSysAve ~ BMI, data = subset(data, Gender == "female"))
abline(female_fit_bmi, col = "red", lwd = 2)
# Add male line of best fit

```

```
male_fit_bmi <- lm(BPSysAve ~ BMI, data = subset(data, Gender == "male"))
abline(male_fit_bmi, col = "blue", lwd = 2)

legend("topleft",
      legend = c("Female", "Male"),
      col = c("red", "blue"),
      pch = 19, lwd = 2, bty = "n")
```

Interaction of BMI and Gender on Blood Pressure

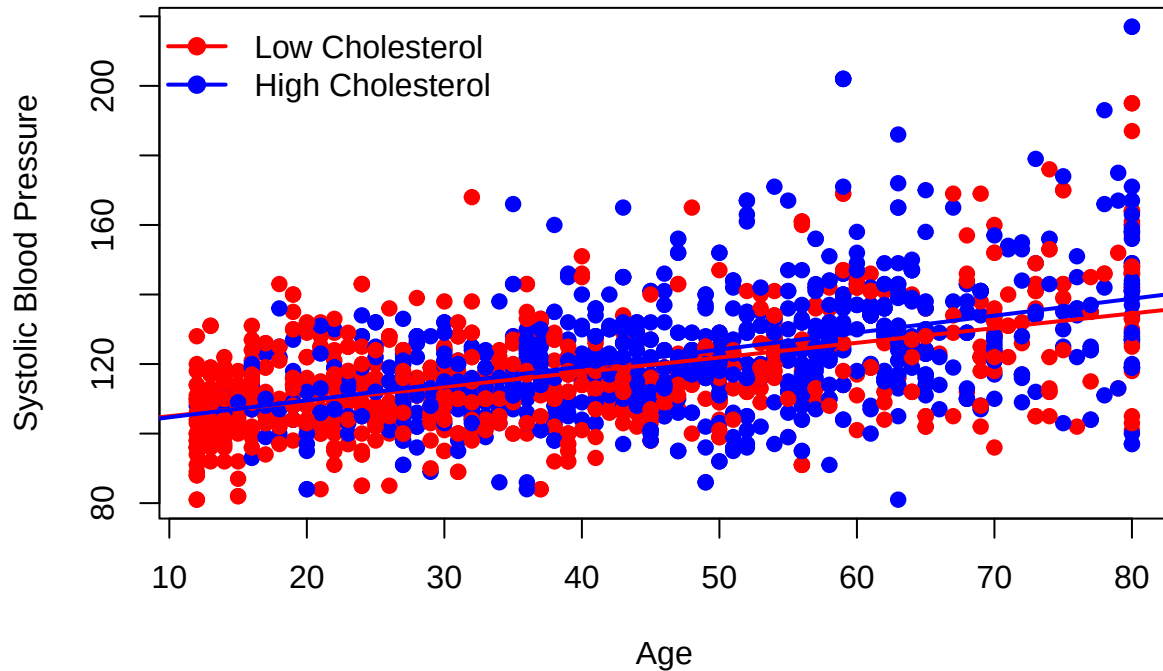


```
# Split TotChol into a high and low category using the median, we will use this to visualize different
median_chol <- median(data$TotChol)
```

```
# Make a scatterplot of blood pressure and cholesterol
plot(data$Age, data$BPSysAve,
      main = "Interaction of Age and Total Cholesterol on Blood Pressure",
      xlab = "Age",
      ylab = "Systolic Blood Pressure",
      pch = 19,
      col = ifelse(data$TotChol < median_chol, "red", "blue")) #colour based on group
# Add low cholesterol line of best fit
low_chol_fit <- lm(BPSysAve ~ Age, data = subset(data, TotChol < median_chol))
abline(low_chol_fit, col = "red", lwd = 2)
# Add high cholesterol line of best fit
high_chol_fit <- lm(BPSysAve ~ Age, data = subset(data, TotChol >= median_chol))
abline(high_chol_fit, col = "blue", lwd = 2)
```

```
legend("topleft",
      legend = c("Low Cholesterol", "High Cholesterol"),
      col = c("red", "blue"),
      pch = 19, lwd = 2, bty = "n")
```

Interaction of Age and Total Cholesterol on Blood Pressure



```
par(mfrow = c(1, 1))
```

Fitting our preliminary model

```
# Fit a model with our predictors, as well as some interesting interaction terms
model = lm(BPSysAve ~ . + Gender:Age + Gender:BMI + Age:TotChol, data=data)
summary(model)
```

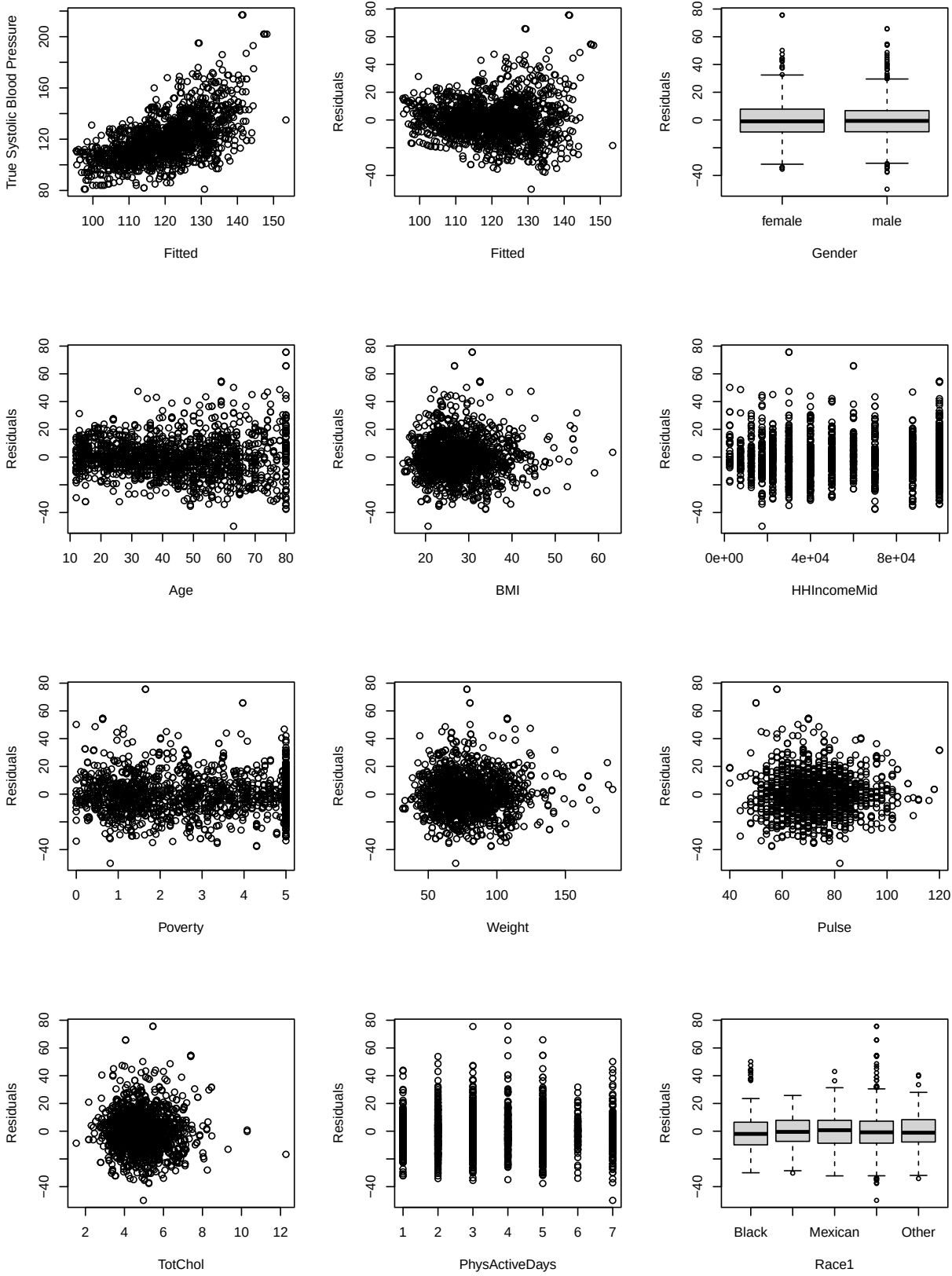
```
##
## Call:
## lm(formula = BPSysAve ~ . + Gender:Age + Gender:BMI + Age:TotChol,
##     data = data)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -49.884  -8.593  -0.827   7.356  75.763
##
```

```

## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)   8.860e+01  6.292e+00  14.083 < 2e-16 ***
## Gendermale    1.511e+01  3.424e+00   4.412 1.09e-05 ***
## Age           2.703e-01  9.934e-02   2.721 0.006574 **
## BMI           1.938e-01  1.557e-01   1.245 0.213490
## HHIncomeMid   1.123e-04  2.328e-05   4.825 1.53e-06 ***
## Poverty       -3.336e+00  4.701e-01  -7.096 1.91e-12 ***
## Weight        4.019e-02  5.077e-02   0.792 0.428760
## Pulse         8.174e-02  3.156e-02   2.590 0.009682 **
## TotChol       -1.812e+00  9.757e-01  -1.857 0.063429 .
## PhysActiveDays -2.732e-01  1.924e-01  -1.420 0.155913
## Race1Hispanic  -6.355e+00  1.925e+00  -3.302 0.000980 ***
## Race1Mexican   -4.583e+00  1.669e+00  -2.747 0.006089 **
## Race1White     -4.419e+00  1.199e+00  -3.687 0.000234 ***
## Race1Other     -3.676e+00  1.689e+00  -2.176 0.029681 *
## PhysActiveYes   7.830e-01  7.810e-01   1.003 0.316187
## TVHrsDay0_to_1_hr 4.234e+00  2.798e+00   1.513 0.130445
## TVHrsDay1_hr    4.575e+00  2.745e+00   1.667 0.095717 .
## TVHrsDay2_hr    5.845e+00  2.702e+00   2.163 0.030698 *
## TVHrsDay3_hr    5.344e+00  2.746e+00   1.947 0.051754 .
## TVHrsDay4_hr    5.138e+00  2.818e+00   1.824 0.068412 .
## TVHrsDayMore_4_hr 4.446e+00  2.784e+00   1.597 0.110494
## Gendermale:Age  -1.561e-01  3.872e-02  -4.033 5.76e-05 ***
## Gendermale:BMI  -1.254e-01  1.162e-01  -1.079 0.280906
## Age:TotChol     6.605e-02  1.952e-02   3.384 0.000731 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 14.19 on 1609 degrees of freedom
## Multiple R-squared:  0.3465, Adjusted R-squared:  0.3372
## F-statistic: 37.1 on 23 and 1609 DF, p-value: < 2.2e-16

```

Residual Plots:



```
vif(model, type="predictor")
```

```
## GVIFs computed for predictors
```

```
##          GVIF Df GVIF^(1/(2*Df)) Interacts With
## Gender      250.302636  5      1.737187      Age, BMI
## Age         33.881422  5      1.422316 Gender, TotChol
## BMI         56.592743  3      1.959417      Gender
## HHIncomeMid  4.778121  1      2.185891      --
## Poverty     4.910053  1      2.215864      --
## Weight      8.907475  1      2.984539      --
## Pulse       1.134785  1      1.065263      --
## TotChol     2.818339  3      1.188499      Age
## PhysActiveDays 1.010401  1      1.005187      --
## Race1       1.327597  4      1.036056      --
## PhysActive   1.196619  1      1.093901      --
## TVHrsDay    1.236799  6      1.017868      --
```

```
##
## Gender      HHIncomeMid, Poverty, Weight, Pulse, TotChol, PhysActiveDays, Race1, PhysA
## Age         BMI, HHIncomeMid, Poverty, Weight, Pulse, PhysActiveDays, Race1, PhysA
## BMI         Age, HHIncomeMid, Poverty, Weight, Pulse, TotChol, PhysActiveDays, Race1, PhysA
## HHIncomeMid Gender, Age, BMI, Poverty, Weight, Pulse, TotChol, PhysActiveDays, Race1, PhysA
## Poverty     Gender, Age, BMI, HHIncomeMid, Weight, Pulse, TotChol, PhysActiveDays, Race1, PhysA
## Weight      Gender, Age, BMI, HHIncomeMid, Poverty, Pulse, TotChol, PhysActiveDays, Race1, PhysA
## Pulse       Gender, Age, BMI, HHIncomeMid, Poverty, Weight, TotChol, PhysActiveDays, Race1, PhysA
## TotChol     Gender, BMI, HHIncomeMid, Poverty, Weight, Pulse, PhysActiveDays, Race1, PhysA
## PhysActiveDays Gender, Age, BMI, HHIncomeMid, Poverty, Weight, Pulse, TotChol, Race1, PhysA
## Race1       Gender, Age, BMI, HHIncomeMid, Poverty, Weight, Pulse, TotChol, PhysActiveDays, PhysA
## PhysActive   Gender, Age, BMI, HHIncomeMid, Poverty, Weight, Pulse, TotChol, PhysActiveDays, PhysA
## TVHrsDay    Gender, Age, BMI, HHIncomeMid, Poverty, Weight, Pulse, TotChol, PhysActiveDays, PhysA
```

VIF seems high for some variables: performing a Partial F Test to see if we can drop Weight and HHIncomeMid

```
full_model = model
reduced_model = lm(BPSysAve ~ Gender + Age + BMI + Poverty + Pulse + TotChol + PhysActiveDays + Race1 +
anova(reduced_model, full_model)
```

```
## Analysis of Variance Table
```

```
##
```

```
## Model 1: BPSysAve ~ Gender + Age + BMI + Poverty + Pulse + TotChol + PhysActiveDays +
```

```
##      Race1 + PhysActive + TVHrsDay + Gender:Age + Gender:BMI +
```

```
##      Age:TotChol
```

```
## Model 2: BPSysAve ~ Gender + Age + BMI + HHIncomeMid + Poverty + Weight +
```

```
##      Pulse + TotChol + PhysActiveDays + Race1 + PhysActive + TVHrsDay +
```

```
##      Gender:Age + Gender:BMI + Age:TotChol
```

```
## Res.Df    RSS Df Sum of Sq    F Pr(>F)
```

```
## 1    1611 328879
```

```
## 2    1609 324045  2    4834.3 12.002 6.7e-06 ***
```

```
## ---
```

```
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
vif(reduced_model, type="predictor")
```

```
## GVIFs computed for predictors
```

```
##          GVIF Df GVIF^(1/(2*Df)) Interacts With
## Gender      29.028056  5      1.400496      Age, BMI
## Age         31.295109  5      1.411067 Gender, TotChol
## BMI         7.665036  3      1.404168      Gender
## Poverty     1.173504  1      1.083284      --
## Pulse       1.133018  1      1.064433      --
## TotChol     2.645012  3      1.175993      Age
## PhysActiveDays 1.009305  1      1.004642      --
## Race1       1.241783  4      1.027438      --
## PhysActive   1.194505  1      1.092934      --
## TVHrsDay     1.211695  6      1.016130      --
##
##                                     Other Predictors
## Gender      Poverty, Pulse, TotChol, PhysActiveDays, Race1, PhysActive, TVHrsDay
## Age         BMI, Poverty, Pulse, PhysActiveDays, Race1, PhysActive, TVHrsDay
## BMI         Age, Poverty, Pulse, TotChol, PhysActiveDays, Race1, PhysActive, TVHrsDay
## Poverty     Gender, Age, BMI, Pulse, TotChol, PhysActiveDays, Race1, PhysActive, TVHrsDay
## Pulse       Gender, Age, BMI, Poverty, TotChol, PhysActiveDays, Race1, PhysActive, TVHrsDay
## TotChol     Gender, BMI, Poverty, Pulse, PhysActiveDays, Race1, PhysActive, TVHrsDay
## PhysActiveDays Gender, Age, BMI, Poverty, Pulse, TotChol, Race1, PhysActive, TVHrsDay
## Race1       Gender, Age, BMI, Poverty, Pulse, TotChol, PhysActiveDays, PhysActive, TVHrsDay
## PhysActive   Gender, Age, BMI, Poverty, Pulse, TotChol, PhysActiveDays, Race1, TVHrsDay
## TVHrsDay     Gender, Age, BMI, Poverty, Pulse, TotChol, PhysActiveDays, Race1, PhysActive
```

Test indicates reduced_model is nearly as good as full model, VIFs are now all < 2

Let's find some outliers we can possible prune

```
#outlier pruning
n <- nrow(data)
p <- length(coef(reduced_model)) - 1

# leverage cutoff
hcut <- 2*((p+1)/n)

# cook's distance cutoff
cookcut <- qf(0.5, p+1, n-p-1)

h_ii <- hatvalues(reduced_model)

# standardized residuals
r_i <- rstandard(reduced_model)

# cook's distance
D_i <- cooks.distance(reduced_model)

# get residuals
residuals_i <- residuals(reduced_model)
```

```

# set a cutoff based on the standardized residuals
negative_residual_cutoff <- -2

# find large negative outliers
treated_outlier_indices <- which(r_i < negative_residual_cutoff)

leverage_outliers <- which(h_ii > hcut)
cooks_outliers <- which(D_i > cookcut)

statistical_influencers <- intersect(leverage_outliers, cooks_outliers)
treated_outliers <- which(r_i < negative_residual_cutoff)

# get all outliers
all_outlier_indices <- unique(union(treated_outliers, statistical_influencers))

cat("--- Final Indices to Consider for Removal ---\n")

## --- Final Indices to Consider for Removal ---

print(all_outlier_indices)

## [1] 3 4 353 373 476 477 698 884 893 931 947 961 977 978 1099
## [16] 1153 1154 1192 1207 1230 1231 1232 1241 1258 1274 1344 1431 1439 1470

data_cleaned <- data[-all_outlier_indices, ]
cat("Original sample size (n):", nrow(data), "\n")

## Original sample size (n): 1633

cat("Number of observations dropped:", length(all_outlier_indices), "\n")

## Number of observations dropped: 29

cat("Cleaned sample size (n_cleaned):", nrow(data_cleaned), "\n")

## Cleaned sample size (n_cleaned): 1604

model_formula <- BPSysAve ~ Gender + Age + BMI + Poverty + Pulse + TotChol + PhysActiveDays + Race1 + P
reduced_model_cleaned <- lm(model_formula, data = data_cleaned)
summary(reduced_model_cleaned)

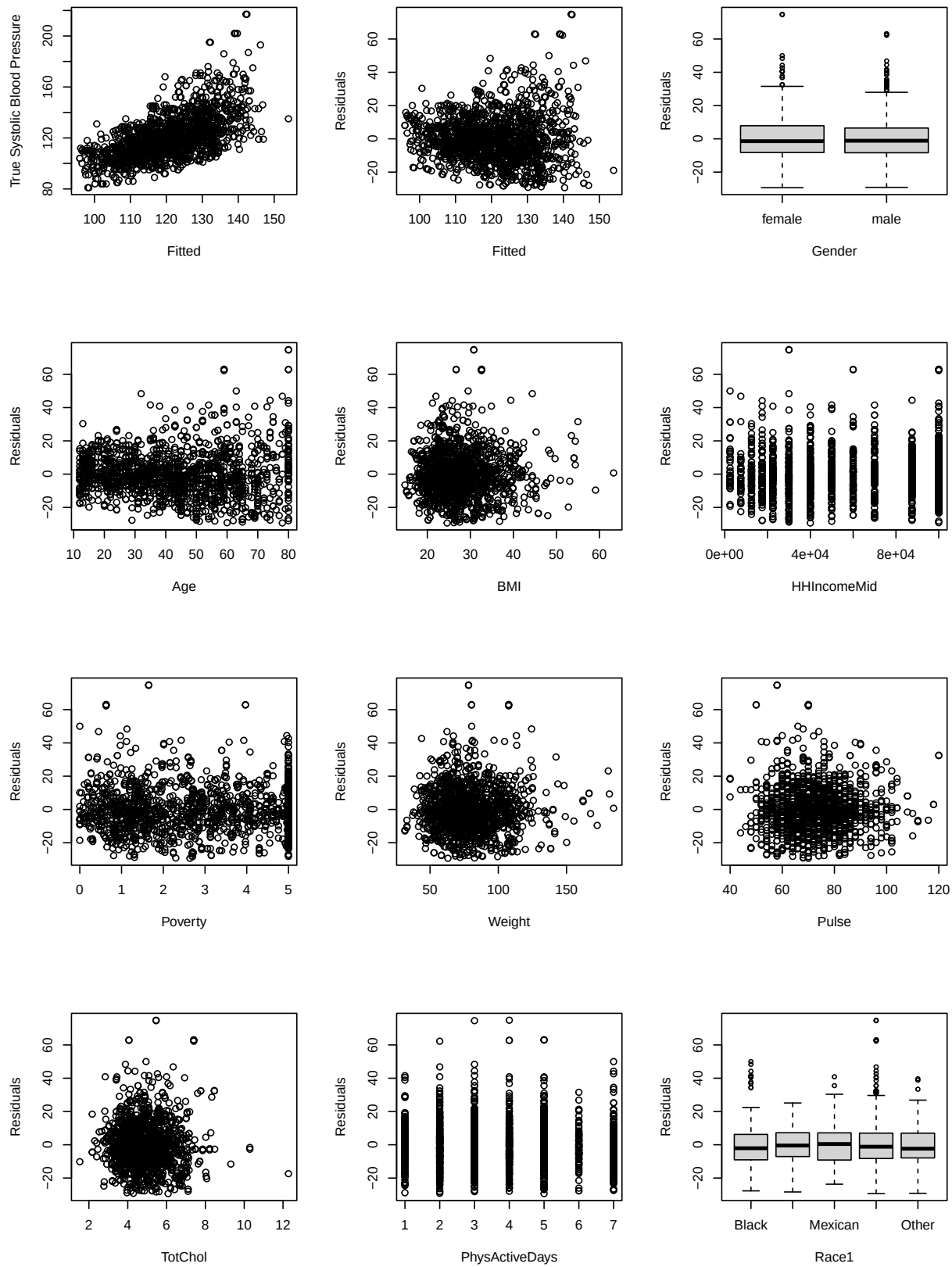
##
## Call:
## lm(formula = model_formula, data = data_cleaned)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -29.346  -8.295  -1.245   6.973  74.848
##

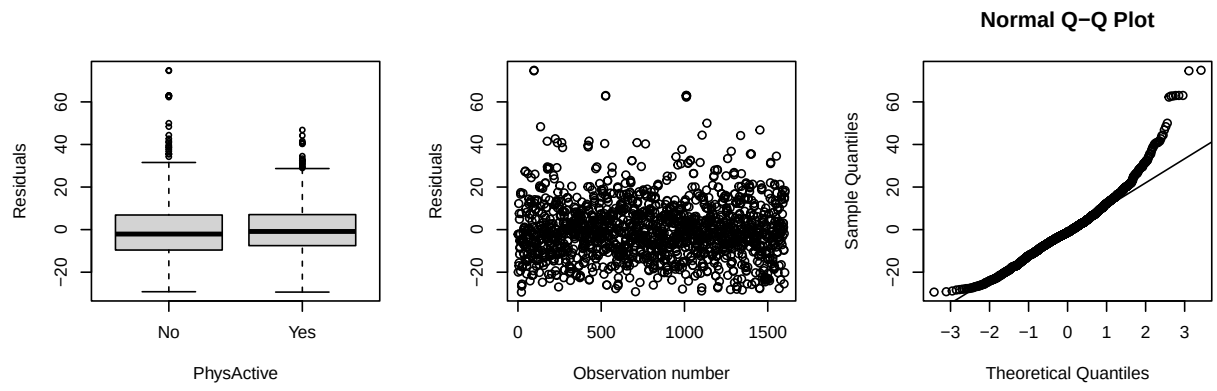
```

```

## Coefficients:
##               Estimate Std. Error t value Pr(>|t|)
## (Intercept)    91.63482     6.09061  15.045 < 2e-16 ***
## Gendermale     14.21552     3.33791   4.259 2.18e-05 ***
## Age            0.24182     0.09655   2.505 0.012361 *
## BMI            0.28338     0.07531   3.763 0.000174 ***
## Poverty       -1.38361     0.22365  -6.186 7.82e-10 ***
## Pulse          0.06648     0.03066   2.168 0.030291 *
## TotChol       -2.03197     0.95052  -2.138 0.032689 *
## PhysActiveDays -0.26193     0.18795  -1.394 0.163616
## Race1Hispanic  -5.74844     1.86040  -3.090 0.002037 **
## Race1Mexican   -4.25239     1.59955  -2.658 0.007928 **
## Race1White     -4.02449     1.16502  -3.454 0.000566 ***
## Race1Other     -2.81940     1.62927  -1.730 0.083741 .
## PhysActiveYes   0.84572     0.75903   1.114 0.265358
## TVHrsDay0_to_1_hr 4.71007     2.70060   1.744 0.081340 .
## TVHrsDay1_hr    5.93081     2.65184   2.236 0.025459 *
## TVHrsDay2_hr    6.48174     2.61033   2.483 0.013127 *
## TVHrsDay3_hr    5.54474     2.65326   2.090 0.036796 *
## TVHrsDay4_hr    6.20022     2.72679   2.274 0.023110 *
## TVHrsDayMore_4_hr 5.22113     2.69413   1.938 0.052806 .
## Gendermale:Age  -0.13928     0.03790  -3.675 0.000246 ***
## Gendermale:BMI  -0.07977     0.11094  -0.719 0.472238
## Age:TotChol     0.06995     0.01900   3.682 0.000239 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 13.71 on 1582 degrees of freedom
## Multiple R-squared:  0.3739, Adjusted R-squared:  0.3656
## F-statistic: 44.99 on 21 and 1582 DF, p-value: < 2.2e-16

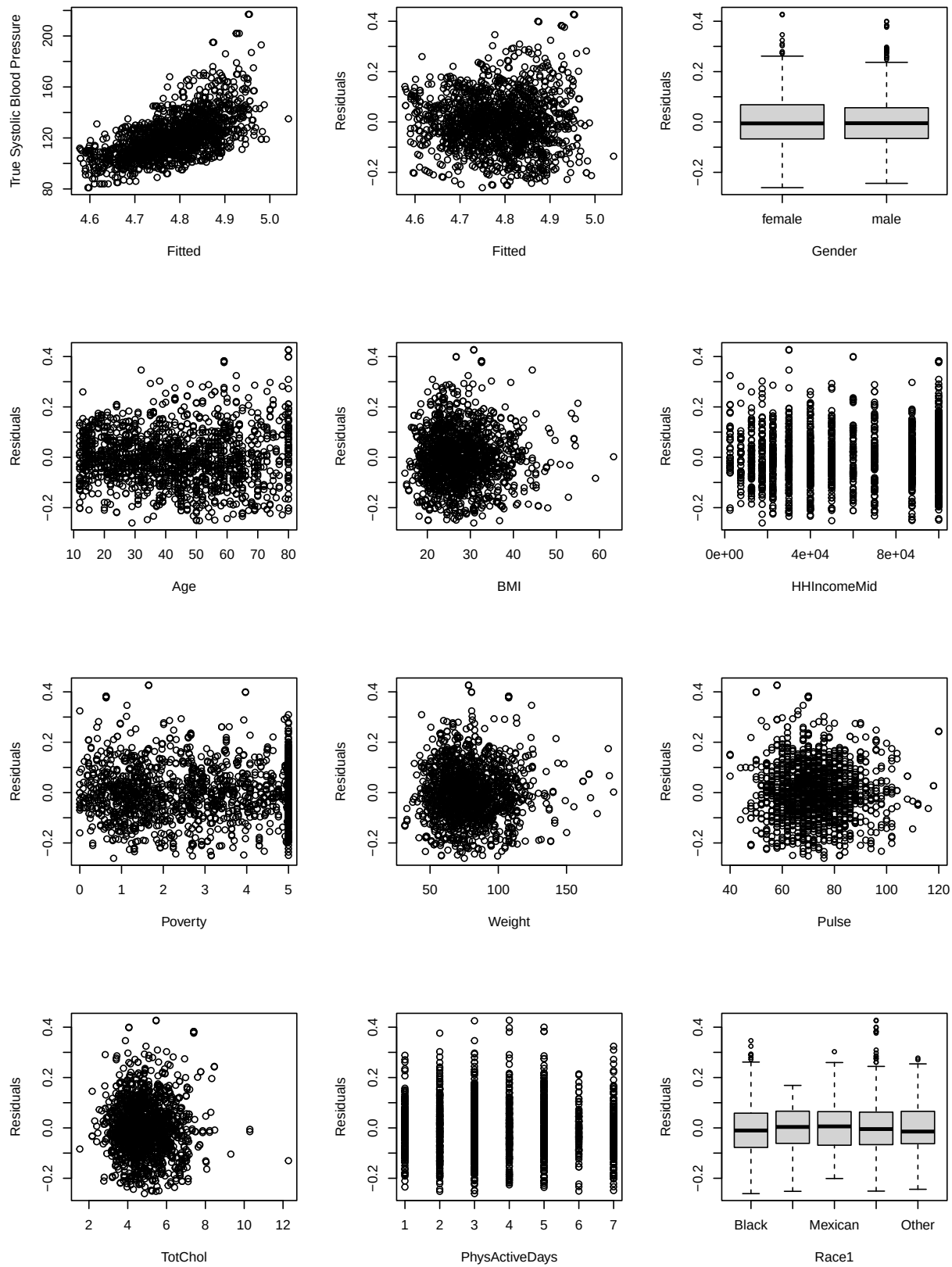
```

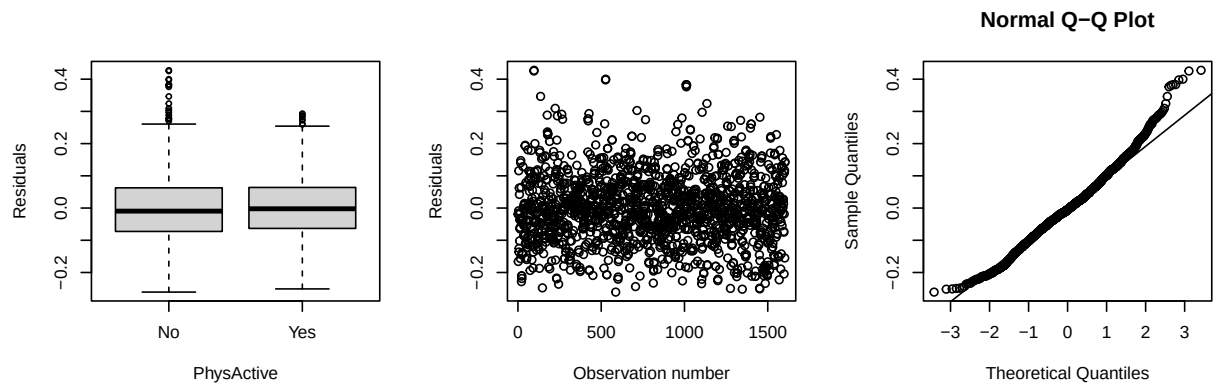




```
logBP = log(data_cleaned$BPSysAve)
reduced_model_cleaned_logged <- lm(logBP ~ Gender + Age + BMI + Poverty + Pulse + TotChol + PhysActiveDays)
summary(reduced_model_cleaned_logged)
```

```
##
## Call:
## lm(formula = logBP ~ Gender + Age + BMI + Poverty + Pulse + TotChol +
##      PhysActiveDays + Race1 + PhysActive + TVHrsDay + Gender:Age +
##      Gender:BMI + Age:TotChol, data = data_cleaned)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -0.26093 -0.06647 -0.00519  0.06366  0.42757
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)    4.5076967   0.0474224   95.054 < 2e-16 ***
## Gendermale      0.1318203   0.0259894    5.072 4.40e-07 ***
## Age             0.0023208   0.0007518    3.087 0.002056 **
## BMI             0.0027499   0.0005863    4.690 2.97e-06 ***
## Poverty        -0.0103703   0.0017414   -5.955 3.19e-09 ***
## Pulse           0.0005362   0.0002387    2.246 0.024855 *
## TotChol        -0.0128485   0.0074009   -1.736 0.082744 .
## PhysActiveDays -0.0022446   0.0014634   -1.534 0.125274
## Race1Hispanic  -0.0465946   0.0144853   -3.217 0.001323 **
## Race1Mexican   -0.0326037   0.0124543   -2.618 0.008932 **
## Race1White     -0.0319672   0.0090710   -3.524 0.000437 ***
## Race1Other     -0.0223786   0.0126857   -1.764 0.077912 .
## PhysActiveYes   0.0091152   0.0059099    1.542 0.123187
## TVHrsDay0_to_1_hr 0.0446706   0.0210273    2.124 0.033791 *
## TVHrsDay1_hr    0.0544098   0.0206476    2.635 0.008492 **
## TVHrsDay2_hr    0.0583044   0.0203244    2.869 0.004176 **
## TVHrsDay3_hr    0.0524392   0.0206586    2.538 0.011232 *
## TVHrsDay4_hr    0.0579457   0.0212311    2.729 0.006418 **
## TVHrsDayMore_4_hr 0.0486248   0.0209769    2.318 0.020575 *
## Gendermale:Age -0.0012913   0.0002951   -4.376 1.29e-05 ***
## Gendermale:BMI -0.0008786   0.0008638   -1.017 0.309284
## Age:TotChol     0.0004878   0.0001479    3.298 0.000995 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.1068 on 1582 degrees of freedom
## Multiple R-squared:  0.3904, Adjusted R-squared:  0.3823
## F-statistic: 48.25 on 21 and 1582 DF, p-value: < 2.2e-16
```





```
final_model = reduced_model_cleaned_logged
vif(final_model, type='predictor')
```

```
## GVIFs computed for predictors
```

##		GVIF	Df	GVIF ^{1/(2*Df)}	Interacts With
##	Gender	28.628360	5	1.398555	Age, BMI
##	Age	30.839089	5	1.408997	Gender, TotChol
##	BMI	7.650876	3	1.403735	Gender
##	Poverty	1.170784	1	1.082028	--
##	Pulse	1.130792	1	1.063387	--
##	TotChol	2.611607	3	1.173504	Age
##	PhysActiveDays	1.010023	1	1.004999	--
##	Race1	1.243297	4	1.027595	--
##	PhysActive	1.188675	1	1.090264	--
##	TVHrsDay	1.212788	6	1.016207	--

##		Other Predictors
##	Gender	Poverty, Pulse, TotChol, PhysActiveDays, Race1, PhysActive, TVHrsDay
##	Age	BMI, Poverty, Pulse, PhysActiveDays, Race1, PhysActive, TVHrsDay
##	BMI	Age, Poverty, Pulse, TotChol, PhysActiveDays, Race1, PhysActive, TVHrsDay
##	Poverty	Gender, Age, BMI, Pulse, TotChol, PhysActiveDays, Race1, PhysActive, TVHrsDay
##	Pulse	Gender, Age, BMI, Poverty, TotChol, PhysActiveDays, Race1, PhysActive, TVHrsDay
##	TotChol	Gender, BMI, Poverty, Pulse, PhysActiveDays, Race1, PhysActive, TVHrsDay
##	PhysActiveDays	Gender, Age, BMI, Poverty, Pulse, TotChol, Race1, PhysActive, TVHrsDay
##	Race1	Gender, Age, BMI, Poverty, Pulse, TotChol, PhysActiveDays, PhysActive, TVHrsDay
##	PhysActive	Gender, Age, BMI, Poverty, Pulse, TotChol, PhysActiveDays, Race1, TVHrsDay
##	TVHrsDay	Gender, Age, BMI, Poverty, Pulse, TotChol, PhysActiveDays, Race1, PhysActive

```
trainLogBP = log(train$BPSysAve)
testLogBP = log(test$BPSysAve)
```

```
model1 = lm(trainLogBP ~ Gender + Age + BMI + Poverty + Pulse + TotChol + PhysActiveDays + Race1 + PhysActive + TVHrsDay, data = train)
model2 = lm(testLogBP ~ Gender + Age + BMI + Poverty + Pulse + TotChol + PhysActiveDays + Race1 + PhysActive + TVHrsDay, data = test)
summary(model1)
```

```
##
## Call:
## lm(formula = trainLogBP ~ Gender + Age + BMI + Poverty + Pulse +
##     TotChol + PhysActiveDays + Race1 + PhysActive + TVHrsDay +
##     Gender:Age + Gender:BMI + Age:TotChol, data = train)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -0.33243 -0.06765 -0.00069  0.06787  0.42578
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)   4.5632118   0.0655516  69.612 < 2e-16 ***
## Gendermale     0.1282518   0.0343256   3.736 0.000198 ***
## Age            0.0008988   0.0010018   0.897 0.369821
## BMI            0.0026191   0.0008260   3.171 0.001568 **
## Poverty       -0.0095400   0.0024000  -3.975 7.55e-05 ***
```

```
## Pulse          0.0005476  0.0003197   1.713 0.087014 .
## TotChol        -0.0227046  0.0099310  -2.286 0.022454 *
## PhysActiveDays -0.0042611  0.0019689  -2.164 0.030691 *
## Race1Hispanic  -0.0334995  0.0197274  -1.698 0.089804 .
## Race1Mexican   -0.0353183  0.0170316  -2.074 0.038369 *
## Race1White     -0.0284193  0.0122349  -2.323 0.020394 *
## Race1Other     -0.0104130  0.0174153  -0.598 0.550031
## PhysActiveYes   0.0042872  0.0081314   0.527 0.598147
## TVHrsDay0_to_1_hr 0.0509719  0.0313291   1.627 0.104063
## TVHrsDay1_hr    0.0549769  0.0308597   1.782 0.075139 .
## TVHrsDay2_hr    0.0573859  0.0305220   1.880 0.060385 .
## TVHrsDay3_hr    0.0661518  0.0308419   2.145 0.032209 *
## TVHrsDay4_hr    0.0589598  0.0315731   1.867 0.062144 .
## TVHrsDayMore_4_hr 0.0504364  0.0312516   1.614 0.106876
## Gendermale:Age  -0.0012315  0.0003956  -3.113 0.001908 **
## Gendermale:BMI  -0.0011549  0.0011589  -0.997 0.319243
## Age:TotChol     0.0007655  0.0001976   3.874 0.000114 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.1139 on 978 degrees of freedom
## Multiple R-squared:  0.3625, Adjusted R-squared:  0.3488
## F-statistic: 26.48 on 21 and 978 DF,  p-value: < 2.2e-16
```

```
summary(model2)
```

```
##
## Call:
## lm(formula = testLogBP ~ Gender + Age + BMI + Poverty + Pulse +
##      TotChol + PhysActiveDays + Race1 + PhysActive + TVHrsDay +
##      Gender:Age + Gender:BMI + Age:TotChol, data = test)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -0.45764 -0.06790 -0.00005  0.05945  0.42352
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)    4.407e+00  7.810e-02  56.431 < 2e-16 ***
## Gendermale      1.373e-01  4.531e-02   3.031 0.002541 **
## Age            4.397e-03  1.286e-03   3.418 0.000673 ***
## BMI            2.958e-03  9.342e-04   3.167 0.001618 **
## Poverty       -1.043e-02  2.850e-03  -3.659 0.000275 ***
## Pulse          6.259e-04  4.090e-04   1.530 0.126428
## TotChol        1.151e-02  1.252e-02   0.919 0.358378
## PhysActiveDays  8.858e-04  2.458e-03   0.360 0.718681
## Race1Hispanic  -8.882e-02  2.396e-02  -3.707 0.000229 ***
## Race1Mexican   -4.462e-02  2.053e-02  -2.174 0.030117 *
## Race1White     -4.832e-02  1.527e-02  -3.164 0.001632 **
## Race1Other     -5.688e-02  2.065e-02  -2.755 0.006048 **
## PhysActiveYes   6.537e-03  9.696e-03   0.674 0.500413
## TVHrsDay0_to_1_hr 3.714e-02  3.159e-02   1.176 0.240106
## TVHrsDay1_hr    3.534e-02  3.080e-02   1.147 0.251661
## TVHrsDay2_hr    5.604e-02  3.017e-02   1.857 0.063728 .
```

```
## TVHrsDay3_hr      2.568e-02  3.080e-02  0.834 0.404651
## TVHrsDay4_hr      3.339e-02  3.172e-02  1.052 0.293016
## TVHrsDayMore_4_hr 2.707e-02  3.140e-02  0.862 0.389002
## Gendermale:Age    -1.498e-03  4.922e-04 -3.044 0.002437 **
## Gendermale:BMI    -4.731e-04  1.485e-03 -0.319 0.750185
## Age:TotChol       -3.624e-05  2.519e-04 -0.144 0.885644
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.1107 on 611 degrees of freedom
## Multiple R-squared:  0.3464, Adjusted R-squared:  0.324
## F-statistic: 15.42 on 21 and 611 DF,  p-value: < 2.2e-16
```

```
vif(model1, type = 'predictor')
```

```
## GVIFs computed for predictors
```

##	GVIF	Df	GVIF ^{1/(2*Df)}	Interacts With
## Gender	29.162993	5	1.401145	Age, BMI
## Age	33.291667	5	1.419821	Gender, TotChol
## BMI	7.898700	3	1.411213	Gender
## Poverty	1.192185	1	1.091872	--
## Pulse	1.146752	1	1.070865	--
## TotChol	2.794939	3	1.186849	Age
## PhysActiveDays	1.016665	1	1.008298	--
## Race1	1.286948	4	1.032037	--
## PhysActive	1.228074	1	1.108185	--
## TVHrsDay	1.301759	6	1.022220	--

##	Other Predictors
## Gender	Poverty, Pulse, TotChol, PhysActiveDays, Race1, PhysActive, TVHrsDay
## Age	BMI, Poverty, Pulse, PhysActiveDays, Race1, PhysActive, TVHrsDay
## BMI	Age, Poverty, Pulse, TotChol, PhysActiveDays, Race1, PhysActive, TVHrsDay
## Poverty	Gender, Age, BMI, Pulse, TotChol, PhysActiveDays, Race1, PhysActive, TVHrsDay
## Pulse	Gender, Age, BMI, Poverty, TotChol, PhysActiveDays, Race1, PhysActive, TVHrsDay
## TotChol	Gender, BMI, Poverty, Pulse, PhysActiveDays, Race1, PhysActive, TVHrsDay
## PhysActiveDays	Gender, Age, BMI, Poverty, Pulse, TotChol, Race1, PhysActive, TVHrsDay
## Race1	Gender, Age, BMI, Poverty, Pulse, TotChol, PhysActiveDays, PhysActive, TVHrsDay
## PhysActive	Gender, Age, BMI, Poverty, Pulse, TotChol, PhysActiveDays, Race1, TVHrsDay
## TVHrsDay	Gender, Age, BMI, Poverty, Pulse, TotChol, PhysActiveDays, Race1, PhysActive

```
vif(model2, type = 'predictor')
```

```
## GVIFs computed for predictors
```

##	GVIF	Df	GVIF ^{1/(2*Df)}	Interacts With
## Gender	32.974447	5	1.418462	Age, BMI
## Age	32.869284	5	1.418009	Gender, TotChol
## BMI	7.928553	3	1.412101	Gender
## Poverty	1.187299	1	1.089632	--
## Pulse	1.145247	1	1.070162	--
## TotChol	2.633779	3	1.175159	Age
## PhysActiveDays	1.032462	1	1.016101	--

## Race1	1.269566	4	1.030284	--
## PhysActive	1.183931	1	1.088086	--
## TVHrsDay	1.250661	6	1.018814	--
##				Other Predictors
## Gender				Poverty, Pulse, TotChol, PhysActiveDays, Race1, PhysActive, TVHrsDay
## Age				BMI, Poverty, Pulse, PhysActiveDays, Race1, PhysActive, TVHrsDay
## BMI				Age, Poverty, Pulse, TotChol, PhysActiveDays, Race1, PhysActive, TVHrsDay
## Poverty				Gender, Age, BMI, Pulse, TotChol, PhysActiveDays, Race1, PhysActive, TVHrsDay
## Pulse				Gender, Age, BMI, Poverty, TotChol, PhysActiveDays, Race1, PhysActive, TVHrsDay
## TotChol				Gender, BMI, Poverty, Pulse, PhysActiveDays, Race1, PhysActive, TVHrsDay
## PhysActiveDays				Gender, Age, BMI, Poverty, Pulse, TotChol, Race1, PhysActive, TVHrsDay
## Race1				Gender, Age, BMI, Poverty, Pulse, TotChol, PhysActiveDays, PhysActive, TVHrsDay
## PhysActive				Gender, Age, BMI, Poverty, Pulse, TotChol, PhysActiveDays, Race1, TVHrsDay
## TVHrsDay				Gender, Age, BMI, Poverty, Pulse, TotChol, PhysActiveDays, Race1, PhysActive

““